

G-CAF

GovDigital Composable Architecture Framework

Volume 1

Introduction & Overview

v2.0

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Chapter 1

What is Composable Architecture?

Reading time: 3 minutes. No technical background required.

The One-Sentence Version

Composable architecture is a way of building government digital services from **modular, reusable building blocks** instead of building everything from scratch every time.

The Problem It Solves

Today, when a government entity needs a new digital service — say, a permit renewal system or a citizen notification service — the typical approach is to build (or buy) a complete, standalone system. This creates three problems:

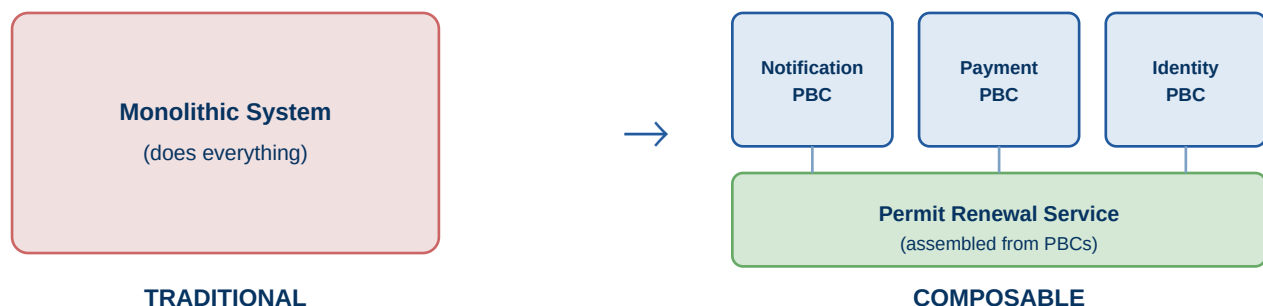
Duplication. Multiple entities end up building the same capabilities independently. Entity A builds a notification engine. Entity B builds a different notification engine. Entity C buys a third one. The government pays three times for the same thing.

Lock-in. Each standalone system comes with its own technology, vendor, data format, and support model. Switching or connecting them later is expensive and slow.

Speed. Building everything from scratch takes time. Citizens wait longer for better services. Entities spend months on infrastructure that already exists elsewhere.

The Idea

Instead of building complete systems, composable architecture breaks digital services into **Packaged Business Capabilities (PBCs)** — self-contained building blocks, each responsible for one specific business function.



Think of it like pre-fabricated, tested, standardised components — walls, roofing systems, electrical units — assembled into different buildings. The components are built once, tested once, and reused many times.

What It Means in Practice

Before (Traditional)	After (Composable)
Each entity builds its own notification system	One government-wide Notification PBC, shared by all entities
Switching vendors requires a full system rewrite	Replace one PBC without affecting others
New services take 12–18 months	New services assembled in weeks from existing PBCs
Knowledge is locked inside each entity	Patterns and PBCs are shared through a central Catalog
No visibility into what already exists	PBC Catalog shows every capability, who owns it, and how to use it

What It Is NOT

Not just APIs. APIs are a communication mechanism. Composable architecture is a governance and design approach that determines what gets built, how it's shared, and who is responsible.

Not just microservices. A PBC can be implemented as microservices, as a SaaS product, or as a vendor solution. The architecture is about business capabilities, not deployment technology.

Not optional reuse. This is not a suggestion to "reuse where convenient." Under G-CAF, the PBC Catalog is the single source of truth, and governance ensures that duplication is avoided by design.

Not one-size-fits-all. Entities retain autonomy over how they assemble PBCs into services. The building blocks are shared; the final products serve each entity's unique needs.

Why It Matters for Abu Dhabi Government

Benefit	Impact
Cost	Shared PBCs reduce duplication. Estimated 30–50% cost reduction.
Speed	New services assembled from existing PBCs. Estimated 50–70% faster delivery.
Citizen Experience	Same PBCs across entities means consistent experience everywhere.
Resilience	Shared PBCs maintained by dedicated teams, monitored 24/7, patched centrally.

How G-CAF Makes It Work

The GovDigital Composable Architecture Framework (G-CAF) is the governance and design framework that makes composable architecture real.

G-CAF Component	What It Does
PBC Catalog	Registry of all building blocks — what exists, who owns it, how to use it
Standard Contracts	Rules each PBC must follow (security, data, integration, performance, operations)
Gateways	Quality checkpoints that every PBC passes before going live
Architecture Principles	17 design principles ensuring PBCs work together
Governance Bodies	ARB (strategic direction) and DDA (day-to-day decisions)

G-CAF Component	What It Does
Reference Architectures	Proven blueprints for common platforms

Chapter 2

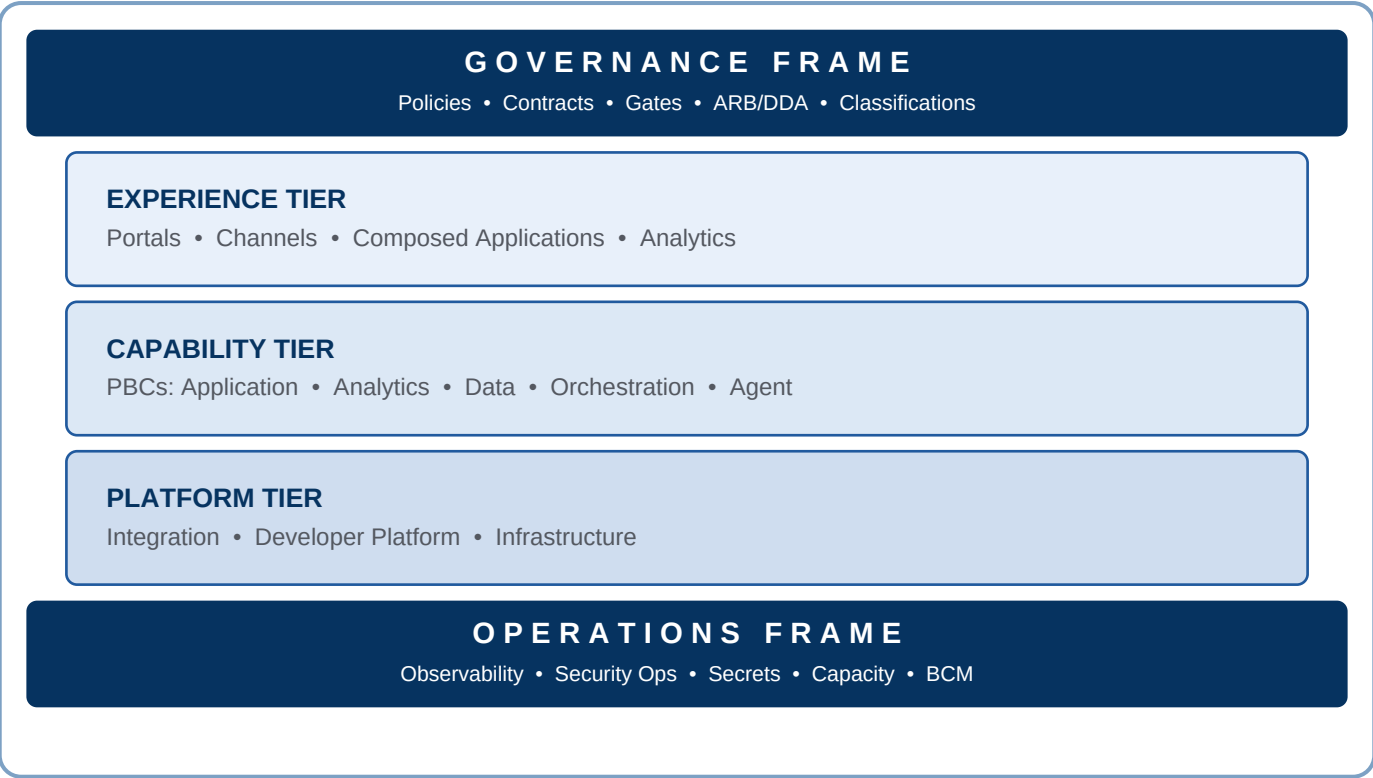
Executive Summary

The GovDigital Composable Architecture Framework (G-CAF) v2.0 establishes the authoritative architectural and governance approach for all government digital services. This framework provides the unified structure that:

- All new digital proposals must align to
- All EA tools and repositories must be structured around
- All digital governance processes must map to

G-CAF enables government entities to design, build, deploy, and operate digital services through modular, reusable Packaged Business Capabilities (PBCs) sourced strategically (build, buy, or outsource) and organized within a clear governance structure.

Framework Structure — 3 Tiers + 2 Frames



3 Tiers (Vertical Stack)

- **Experience Tier** — Where users interact (Portals, Channels, Analytics)
- **Capability Tier** — What we deliver (PBCs, Domains, Agents)
- **Platform Tier** — What we build on (Integration, Developer, Infrastructure)

2 Frames (Horizontal Wrap)

- **Governance Frame** — Design-time policies that shape what we build

- **Operations Frame** — Runtime policies that ensure systems run reliably

Six PBC Types

PBC Type	Description
Application PBCs	Business entity lifecycle management
Analytics PBCs	Insight, predictions, AI services (consume Data Fabric)
Data PBCs	Reference data, lookups, search (operate on Data Fabric)
Orchestration PBCs	Multi-PBC process coordination
Digital Twin PBCs	Physical asset representation
Agent PBCs	Autonomous AI agents with reasoning, tools, memory, and guardrails

Enterprise Catalogs

Catalog	Purpose
PBC Catalog	All registered PBCs with lifecycle status
Technology Standards Catalog	Approved technologies with Strategic/Tactical/Containment/Retired lifecycle
Data Asset Catalog	Data entities, Source of Truth assignments, quality tiers
Integration Asset Catalog	APIs, Events, Schemas with consumer contracts

Seventeen Architecture Principles

Core Principles (ARC-001–010): API-First, Cloud-Native, Security by Design, Privacy by Design, Zero Trust, Accessibility First, Open Standards, Reuse Before Build, Loosely Coupled, Design for Failure

Evolvability Principles (ARC-011–013): Interface Stability, Tolerant Evolution, Versioned Compliance

Operational Excellence (ARC-014–017): Citizen Experience First, Data Quality First, Observable by Default, Sustainable by Design

Core Value Proposition

Metric	Value
Service Delivery Speed	50–70% faster through reusable PBCs
Cost Reduction	30–50% via shared PBCs and optimal sourcing
Governance Coverage	100% through PBC Catalog and DDA oversight
Change Impact	Bounded through versioned contracts and interface stability
AI Readiness	Agent PBC governance with MCP/A2A standards

Metric	Value
Accessibility	WCAG compliance with RTL/Arabic support

Who Should Read This

Role	Recommended Focus
Executives	Introduction, Governance Frame, value summaries
Architects	All sections for complete framework understanding
Platform Teams	Tiers, Environment Standards, Reference Architectures
Entity Leaders	Governance, Operational Models, Sourcing Strategy
Security/Data Teams	Standard Contracts (Security, Data), Agent Governance
AI/Agent Teams	Agent PBC Type, Agent Governance Contract
Procurement	Sourcing, DDA Technical Committee process

Chapter 3

Framework Identity

This chapter is the **single source of truth** for G-CAF naming, acronym expansion, key messaging, and brand terminology.

Canonical Identity

Field	Value
Acronym	G-CAF
Full Name	GovDigital Composable Architecture Framework
Short Name	G-CAF
Version	2.0
Architecture Model	3 Tiers + 2 Frames
Owner	Enterprise Architecture Office
Governing Entity	Department of Government Enablement (DGE), Abu Dhabi

Naming Rules

MUST:

- Always expand the acronym on first use: "GovDigital Composable Architecture Framework (G-CAF)"
- Use "G-CAF" (with hyphen) for all subsequent references
- Include "GovDigital" as one word (capital G, capital D, no space)

MUST NOT:

- Never expand as "Government Cloud Adoption Framework"
- Never use "GCAF" (without hyphen)
- Never use "Gov Digital" (with space)

Elevator Pitch

*G-CAF organizes composable government architecture into **three tiers** — Experience, Capability, and Platform — wrapped by **two frames**: Governance (which shapes what we build) and Operations (which ensures it runs). The **Packaged Business Capability (PBC)** remains the central unit of delivery and governance.*

Key Messages by Audience

Audience	Key Message
Executive / Leadership	G-CAF v2.0 simplifies our architecture model from 9 layers to 5 concepts, making it easier to communicate.
Architects	The 3T+2F model correctly represents cross-cutting concerns as frames and stacking concerns as tiers.
Developers	Nothing changes in how you build. The framework just has clearer names now.
External / Vendors	GovDigital CAF aligns with Gartner's composable platform model.

Common Misnamings

Incorrect	Correct	Root Cause
Government Cloud Adoption Framework	GovDigital Composable Architecture Framework	Azure/AWS CAF confusion
GCAF	G-CAF	Missing hyphen
Gov Digital	GovDigital	Incorrect spacing

Chapter 4

Purpose & Scope

G-CAF serves as the **single reference framework** for:

Area	Description
PBC Governance	All PBCs (new and existing) are governed through the DDA and PBC Catalog
EA Tools and Repositories	Architecture tools, catalogs, and repositories must be structured according to G-CAF
Digital Governance	All governance processes center on PBCs as the unit of value
Procurement	All digital procurements for PBCs follow DDA Technical Committee process
PBC Lifecycle	All PBCs pass mandatory PBC Gateways from initiation through retirement
Standards Compliance	All PBCs must declare compliance with specific cross-cutting standard contract versions
Interface Contracts	All PBCs must publish stable interface contracts separate from internal implementation
Schema Evolution	All PBC APIs must follow schema evolution rules to prevent breaking changes

Scope

G-CAF applies to:

- All government digital services and platforms (new and existing)
- All Packaged Business Capabilities (PBCs) regardless of sourcing
- Shared capabilities and infrastructure
- Entity-specific applications and services
- Cross-entity integration and data exchange
- Custom-built PBCs, vendor-sourced PBCs, and outsourced PBC implementation

Framework Authority

G-CAF is established under the authority of **Executive Circular DGO-GOVDIGITAL/EA/2025/001**, which activates:

- The Architecture Review Board (ARB) as strategic authority for Capability Portfolio and Standard Contract Versions
- The Digital Design Authority (DDA) as PBC governance body and Technical Committee
- Mandatory PBC Gateway reviews for all PBCs

Compliance is mandatory. PBCs that do not align to G-CAF or bypass EA governance may not be registered in the PBC Catalog and may face withdrawal of GovDigital endorsement.

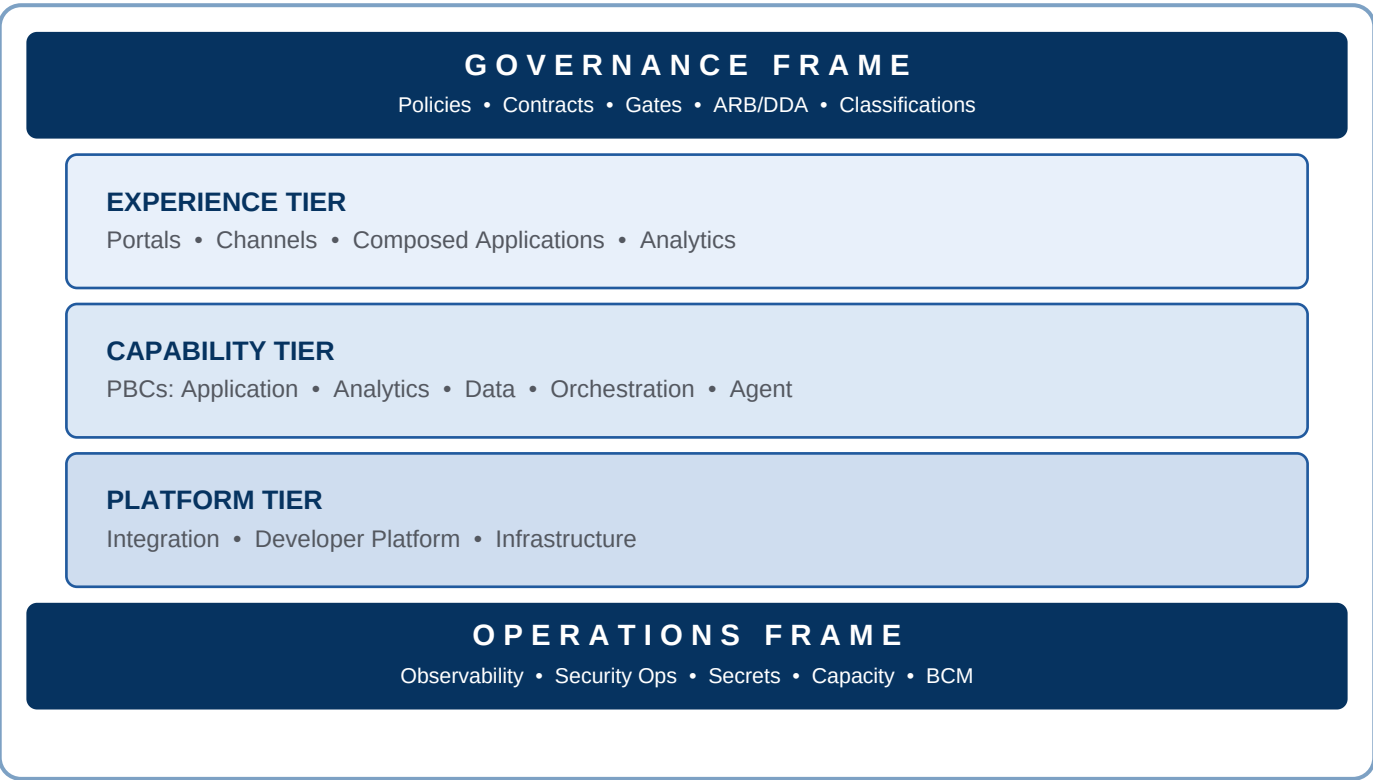
Evolvability Principles (NS Alignment)

Principle	Description	Mechanism
Separation of Concerns	Each component handles one concern	3T+2F + orthogonal classification
Action Version Transparency	PBC changes don't break callers	Interface Contracts + API versioning
Data Version Transparency	Schema changes don't break consumers	Schema evolution + tolerant reader
Separation of State	PBC state is independent	Clear ownership + lifecycle states

Chapter 5

Framework Overview — 3 Tiers + 2 Frames

G-CAF v2.0 organizes composable government architecture into **three tiers** wrapped by **two frames**. The Packaged Business Capability (PBC) remains the central unit of delivery and governance.



Why 3 Tiers + 2 Frames?

The previous 9-layer model (Layers 0–8) created cognitive friction — misleading numbering, confusing topology, and arbitrary grouping. The solution: **Tiers** are vertical — they stack. **Frames** are horizontal — they wrap around everything.

Experience Tier

Purpose: Where users interact with government services.

Component	Description
Portals	G2C (Citizen), G2B (Business), G2G (Government), G2E (Employee)
Channels	Web, Mobile, Kiosk, API, Voice, Chatbot
Composed Applications	Role-based assemblies of PBC capabilities
Analytics Experiences	BI dashboards, reports, semantic layer

Capability Tier

Purpose: What we deliver — the building blocks of government services.

Component	Description
PBCs	Application, Analytics, Data, Orchestration, Digital Twin, Agent
Domains	Bounded contexts, aggregates, ownership boundaries
Domain Events	Business events that cross boundaries
Ubiquitous Language	Shared vocabulary within domains

Platform Tier

Purpose: What teams build on — shared infrastructure and services.

Sub-Tier	Components
Integration	API Gateway, Event Bus, Schema Registry, Data Fabric, Orchestration
Developer	IDP, Pipelines, Golden Paths, Composition Platform, Tooling Standards
Infrastructure	DPI (Digital ID, Payments, Registries), Cloud (Gov Cloud, SaaS, Hybrid)

Governance Frame (Design-Time)

Purpose: Shape what gets built before it's built. Governance is enabling, not blocking.

Component	Description
Governance Bodies	ARB (Architecture Review Board), DDA (Digital Design Authority)
Standard Contracts	Security, Data, Integration, NFR, Operations
Gateways	Initiation, Design, Go-Live, Retirement
Classifications	Criticality, Sourcing, Stability, Maturity, PBC Types
Principles	17 architecture principles governing all decisions

Operations Frame (Runtime)

Purpose: Ensure systems run reliably in production. Operations observes all tiers.

Component	Description
Security Operations	SOC, SIEM, Incident Response
Secrets Management	Key Vault, Rotation, Access Control
Observability	Metrics, Logs, Traces, Alerting
Capacity Management	Sizing, Scaling, Performance

Component	Description
Business Continuity	DR, Backup, RTO/RPO

Industry Alignment

This model aligns with Gartner's Composable Business Platform research:

Gartner Term	G-CAF Equivalent
Experience Platforms	Experience Tier
Core Platforms	Capability Tier
Foundational Platforms	Platform Tier
Fusion Teams	PBC Teams / Domain Teams
Packaged Business Capabilities	PBCs

Migration: v1.x Layers to v2.0 Model

v1.x Layer	v2.0 Location
Governance Frame	Governance Frame (now explicitly cross-cutting)
Experience Tier	Experience Tier (unchanged)
Capability Tier	Capability Tier (unchanged)
Platform Tier (GovDX)	Platform Tier → Integration
Platform Tier (Data Fabric)	Platform Tier → Integration
Platform Tier (IDP)	Platform Tier → Developer
Platform Tier (DPI)	Platform Tier → Infrastructure
Platform Tier (Cloud)	Platform Tier → Infrastructure
Operations Frame	Operations Frame (now explicitly cross-cutting)

Chapter 6

How to Use This Framework

Usage by Role

If You Are...	Use G-CAF To...
Proposing a new PBC	Check PBC Catalog for reuse; determine classification; prepare for PBC Gateways
Building/Buying a PBC	Follow DDA review; declare standard contract versions; define interface contracts
Evaluating architecture	Validate PBC alignment to reference architectures and patterns
Configuring EA tools	Structure PBC Catalog according to G-CAF layers, domains, and classifications
Conducting procurement	Follow DDA Technical Committee process for vendor PBCs
Managing PBC lifecycle	Pass PBC Gateways; maintain catalog registration; plan retirement
Evolving standards	Publish new standard contract versions with migration windows
Changing PBC APIs	Follow schema evolution rules; maintain interface contract stability

Common Tasks — Quick Start

"I want to propose a new PBC"

1. Check PBC Catalog for existing PBCs (avoid duplication)
2. Determine Classification: Criticality, Sourcing, PBC Type
3. Prepare for Initiation Gateway

"I need to understand governance"

1. Read about the ARB — strategic decisions
2. Read about the DDA — PBC-level governance
3. Review Gateways — lifecycle checkpoints

"I'm integrating a PBC"

1. Review Integration Contract — API/event standards
2. Review Interface Framework — contract structure
3. Review GovDX Integration Platform specifications

Chapter 7

Document Metadata

Field	Value
Document	GovDigital Composable Architecture Framework (G-CAF)
Version	2.0
Status	Approved for Implementation
Effective Date	December 2025
Next Review	June 2026
Classification	Government — For Official Use
Executive Circular	DGO-GOVDIGITAL/EA/2025/001
Owner	Enterprise Architecture Office
Contact	govdigital.ea@dge.gov.ae

Revision History

Version	Date	Summary
2.0	January 2026	3 Tiers + 2 Frames model replacing 9-layer architecture
1.1	December 2025	NS-Compliant Operational Governance — new single-change-driver modules
1.0	November 2025	Initial release — comprehensive G-CAF framework

Chapter 8

Volume Guide — What to Read Next

The G-CAF documentation is organized into nine volumes. This document is Volume 1. The complete set covers all aspects of the framework:

Volume	Title	Audience	Status
Vol 1	Introduction & Overview	Everyone	This document
Vol 2	Framework Core (Tiers, Frames, Principles)	Architects	Planned
Vol 3	Contracts & Governance	Architects, Governance	Planned
Vol 4	Business Drivers & Strategy	Leadership, Strategy	Planned
Vol 5	Reference Architectures	Solution Architects	Planned
Vol 6	Implementation Guide	Program Managers	Planned
Vol 7	Assessments & Compliance	Auditors, Compliance	Planned
Vol 8	Templates & Catalogs	Procurement, Teams	Planned
Vol 9	Theories & Foundations	Deep-dive readers	Planned

Recommended Reading Paths

You Are...	Read These Volumes
An executive who needs the big picture	Vol 1 → Vol 4 (Business Drivers)
An architect designing PBCs	Vol 1 → Vol 2 → Vol 3 → Vol 5
A program manager implementing G-CAF	Vol 1 → Vol 6 → Vol 7
A procurement officer evaluating vendors	Vol 1 → Vol 8
Someone wanting deep theory	Vol 1 → Vol 9 → Vol 2